Computation and Visualization

Project 2

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Data Visualization Dashboard for Suffolk County, Massachusetts

This project involves the creation of an interactive data visualization dashboard that analyzes multiple socio-economic and regional variables at the zip code level for Suffolk County, Massachusetts. The dashboard showcases the overall suitability of Suffolk County zip codes based on five key variables: Median Income, Educational Attainment, Population Density, Housing Prices, and Poverty Percentage. These variables were chosen because they represent critical socio-economic factors that significantly influence individuals' decisions regarding the desirability and suitability of a region. The data for these variables were sourced from credible platforms, such as the U.S. Census Bureau for demographic and economic data and Zillow for housing prices.

Score Calculation Explanation:

A **composite score** is generated for each zip code to quantify its overall desirability. The calculation follows a structured methodology:

1. **Normalization**: Each variable is normalized to ensure an equal contribution to the composite score. Variables such as Median Income, Education, and Population Density are directly proportional to desirability (higher values indicate better living conditions). In contrast, variables like Housing Prices and Poverty are inversely proportional (lower values are more desirable).

2. **Weighting System**: Users can assign weights (ranging from 0–100) to each variable to reflect their priorities and preferences. For instance, assigning a higher weight to the Median Income emphasizes economic prosperity while prioritizing Housing Prices highlights affordability as a critical factor.

Tableau Public Dashboard Link:
https://public.tableau.com/shared/KJ3Y33TB7?:display_count=n&:origin=viz_share_link

References

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